Teacher Reflection Lab: Designing a School-Based Practice Space for Collaborative Teacher Reflection

Ruiqi & Yuzhi

Introduction

Teaching requires constant in-the-moment judgment under complex and often unpredictable conditions. Yet teachers do not always have structured opportunities to slow down, revisit difficult moments, and learn from one another in low-stakes settings. Research on practice-based teacher education argues that teachers need spaces to rehearse, reflect on, and discuss important decisions in teaching, rather than relying solely on theory-heavy coursework or the uneven experiences of daily classroom practice (Reich et al., 2018). Teacher practice spaces are valuable because they combine the authenticity of real teaching with the support and reflection typically found in professional learning settings. Rather than trying to replicate the full complexity of classroom life, such spaces can focus attention on specific decisions, dilemmas, and dispositions that matter for teaching.

Our group's project was designed in response to this need. It helps schools create a dedicated space where teachers can collaboratively reflect on real classroom challenges, share strategies, and build a growing archive of teaching experiences. Instead of producing a fixed digital platform, we designed a manual that schools can adapt to their own space, schedule, staffing patterns, and professional development goals. The manual explains how to set up the room, organize materials, facilitate discussion, and archive cases so that the activity can become a sustainable part of regular teacher collaboration.

The central learning goal of our design is to support teachers in developing more reflective, empathetic, and community-based approaches to classroom problem solving. More specifically, we aimed to create a structure that helps teachers analyze classroom dilemmas from multiple perspectives, especially by considering both peer responses and students' contexts. Our teaching goal was not simply to help teachers "solve" isolated problems, but to help them build habits of collective reflection, perspective-taking, and professional knowledge sharing over time. In this sense, the Teacher Reflection Lab functions as a teacher practice space: it is a low-stakes, structured environment where teachers can revisit problems of practice, learn from colleagues, and generate a shared repertoire of responses.

The intended users of this product are in-service teachers working in schools that already have some form of professional collaboration time, such as grade-level meetings, department meetings, or professional development sessions. The manual is especially suitable for K–12 school settings where teachers encounter recurring classroom challenges but may lack structured opportunities to discuss them in depth with colleagues. These challenges may include student emotional responses, conflict, participation barriers, classroom management issues, or other dilemmas that are difficult to address through generic training alone.

The intended deployment context is a school-based room or meeting space that can be temporarily or permanently organized as a reflection lab. Our design relies mainly on simple materials such as printed case sheets, sticky notes, folders, wall labels, and storage space. A projector can support the session, but it is optional. The room is organized so that teachers can first move around and examine posted cases, then gather for whole-group discussion, and finally preserve the cases and reflections in an archive for future use. Because the system is lightweight and reusable, it can be integrated into existing school routines rather than requiring schools to adopt expensive infrastructure or an entirely new platform.

We also designed the toolkit with adaptability in mind. Different schools have different professional cultures, physical environments, and staff capacities. For that reason, the manual does not prescribe a rigid script. Schools can change the room layout, categories of cases, facilitation process, and meeting frequency. This flexibility allows the Reflection Lab to respond to local needs rather than imposing a one-size-fits-all system.

Design Rationale and How the Product Supports Learning

Our design was informed by research on practice-based teacher education. It is suggested that teacher practice spaces are useful because they allow teachers to rehearse for and reflect on important decisions in teaching in environments that are lower-stakes than live classrooms but more concrete than abstract discussion alone (Reich et al., 2018).

Our project supports learning by structuring teacher reflection around authentic cases contributed by teachers themselves. Before each session, teachers anonymously submit a recent classroom challenge and the response they attempted. This creates a pool of real problems of practice rather than invented examples. During the session, teachers engage in two connected learning moves. First, during case exploration, they respond to peers' cases using two different kinds of sticky notes: one for possible teacher strategies and one for what the student might be experiencing. This dual structure was intentional. It pushes teachers beyond solution-oriented thinking alone and invites them to consider both instructional action and student perspective. Second, during group discussion, teachers examine patterns across cases, compare approaches, and surface broader themes in their school context. This makes the activity both individual and collective: each case is specific, but discussion helps teachers identify recurring challenges and shared professional concerns.

The product's learning objective is to let teachers strengthen their ability to collaboratively interpret classroom dilemmas, consider multiple perspectives, and generate more reflective, empathetic, and context-sensitive responses to common problems of practice. Reich et al. (2018) argue that effective practice spaces can function like "drills" rather than full "scrimmages," focusing attention on useful skills and dispositions that can later be reintegrated into real teaching. Our Reflection Lab follows this logic. It does not simulate a whole class period. Instead, it isolates and foregrounds one essential dimension of teaching: making sense of difficult classroom moments through reflection, empathy, and shared professional dialogue.

Our design also draws on insights from *Teacher Moments*. Participants in *Teacher Moments* perceived the experience as authentic and cognitively challenging, and the platform created a shared basis for debriefing and feedback (Owho-Ovuakporie et al., n.d.). We were especially influenced by the idea that learning is strengthened when teachers share a common experience to analyze together. In our final design, the “common experience” is not a video simulation but the collective examination of real cases posted in the room. The archive extends this further: over time, teachers are not only discussing isolated events, but also building a reusable institutional memory of classroom challenges and responses.

Another key design consideration was cognitive load. In digital systems, it is easy to add more steps, screens, and interactions in ways that feel innovative but actually slow users down. We realized that for practicing teachers, efficiency matters. A reflection tool must not create unnecessary friction or demand excessive time and attention. Our shift away from a centralized digital experience and toward a flexible physical toolkit reflects this concern. The final manual reduces technical complexity while preserving the most meaningful aspects of the experience: authentic cases, structured reflection, collaborative discussion, and archiving.

Finally, our design leaves room for AI in a limited and purposeful role. In the manual, the facilitator may optionally use AI to identify themes across submitted cases and generate a discussion agenda. This idea is informed by recent research arguing that AI in teacher learning should be guided by educator expertise rather than replacing it. Barno et al. (2024) emphasize that generative feedback systems are most educationally useful when teacher educators can edit, endorse, and shape the feedback process. In our design, AI does not evaluate teachers or determine solutions. It only supports pattern recognition and facilitation, while the core learning remains teacher-driven and community-based.

Design Process and Iteration through Playtest

At the beginning of the project, we envisioned a ‘Classroom Chaos Exhibition’, a physical pop-up experience similar to a museum installation. The goal was to build empathy toward teachers by showing that teaching involves constant decision-making in unpredictable situations. We wanted users to move through an environment that displayed classroom dilemmas and invited reflection on the pressures teachers face. Since we could not build the physical exhibition, we used Vibe Coding to create a digital prototype to demonstrate the concept and overall experience. But as we developed it, we realized it didn’t have to stay as a prototype. It could become a fully interactive online experience where people can explore and engage with it directly, rather than just a physical exhibition.

Our first playtest focused on whether the experience flow was clear and understandable. Overall, users followed the structure and responded positively to the comparison between AI-generated suggestions and human responses, which helped highlight different approaches to classroom situations. At the same time, feedback noted that the visual style felt limited and somewhat game-like, which diminished the sense of realism and immersion. Based on this, we refined both the interface and interaction. We explored various visual approaches to enhance the environment's quality and simplified navigation by adding filters and reducing unnecessary steps. Instead of moving through every stage manually, users could select a scenario and directly enter a complete experience centered on that case.

In the second playtest, these changes improved clarity and usability. The system felt more coherent and easier to navigate, and users had fewer surface-level suggestions. However, this round revealed a more important issue. Even with a smoother flow, the experience felt more like going through a sequence than actively engaging. Compared to a physical exhibition, the digital version lacked a sense of participation and decision-making. In addition, one of our play testers, who is a teacher, noted that if this were used as a real platform, the structure could still feel inefficient. The number of steps and interactions made it harder to quickly access relevant content, potentially creating unnecessary cognitive effort in practice.

These insights led us to rethink the project's direction. While we had explored turning the design into a fully digital platform, playtesting suggested that this approach did not best support engagement or usability. It also did not fully align with our original goal of building teacher community and supporting reflection in a more natural setting. As a result, we returned to our initial intention and reframed the project from a different angle. Instead of focusing on creating a centralized digital experience, we began exploring how this exhibition could become more accessible and adaptable across different contexts. Rather than being tied to a fixed location or a single online platform, the experience could be implemented by different schools using simple materials within their own spaces. This led to a shift toward a structured, adaptable toolkit. Instead of presenting the project as a standalone digital product, we positioned it as a guide that helps schools set up similar activities locally. This also allows the experience to be tailored to different needs, making it more flexible and relevant to specific teaching environments. By reducing complexity in the digital layer and focusing on real-world applications, the final design better supports engagement, lowers unnecessary cognitive load, and more directly meets the goal of fostering empathy, reflection, and community among teachers.

Improvements for Future Revision

If we were to revise the project further, we would make several improvements. First, we would conduct longer-term field testing in an actual school setting. Our current evidence comes mainly from prototype playtesting and conceptual refinement. Reich et al. (2018) note that small playtests are useful for iterative refinement, but stronger evidence of impact requires a deeper study of how teacher behavior and reflection practices change over time. Our next step would be to implement the Teacher Reflection Lab over several weeks or months in one or more schools in order to study participation patterns, facilitator needs, teacher perceptions, and the kinds of insights that emerge in the archive over time.

Second, we would strengthen the archive system. Right now, the archive is conceptually important, but it could be more fully designed. In a future version, we would develop clearer tagging systems, retrieval structures, and protocols for revisiting older cases. This would make the archive not just a storage system, but a more active source of collective professional memory.

Third, we would create facilitator supports. Since the quality of discussion may depend heavily on facilitation, a revised manual could include facilitator scripts, discussion prompts, example agendas, and norms for handling sensitive or emotionally difficult cases. This would improve consistency while still preserving local flexibility.

Fourth, we would explore a hybrid version of the toolkit. Our process taught us that a fully digital platform was not the right final direction, but some digital support could still be useful. For example, an online submission form, automated case organization, searchable digital archives, or optional AI-assisted theme clustering could complement the physical space without replacing it.

Finally, we would gather more direct evidence about whether the Reflection Lab helps teachers develop specific dispositions or practices, such as empathy, reflective interpretation, or collaborative problem framing. This might involve pre- and post-reflection tasks, interviews, or analysis of how teachers' written responses evolve across multiple sessions.

References

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